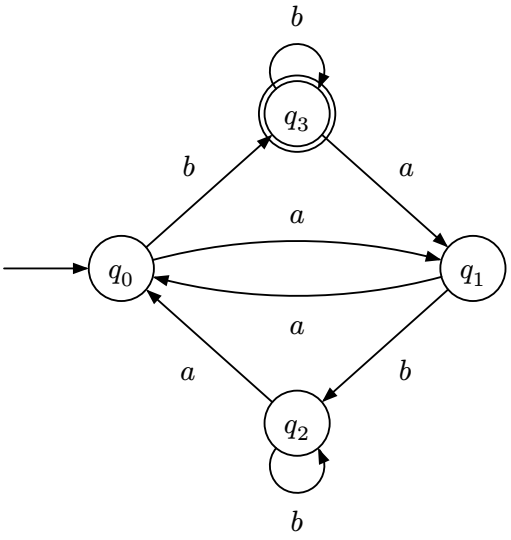
CMSC 331: Homework 5

Carson Kegley
Professor Chang

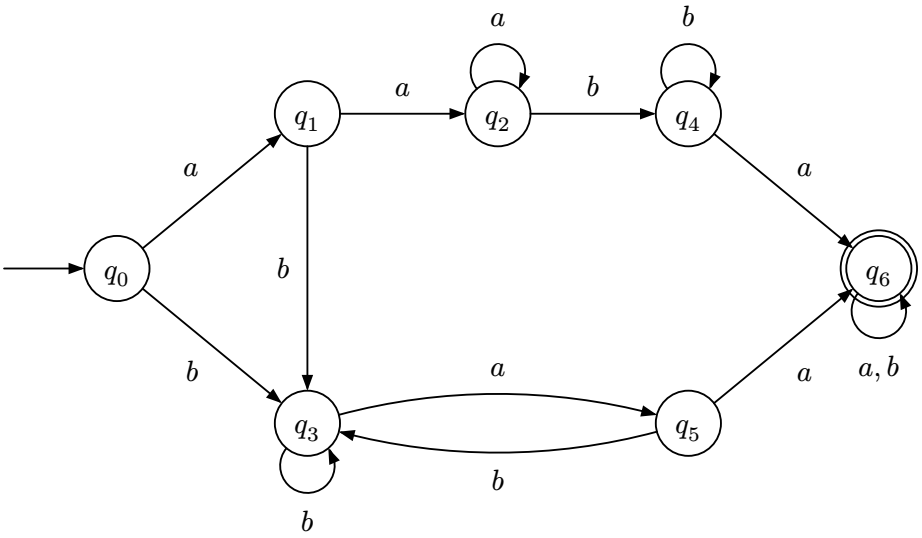
Problem 1: Create DFAs for the following three sets

1. A = { w | w is a string over {a, b}, has an even number of a's and ends with b }

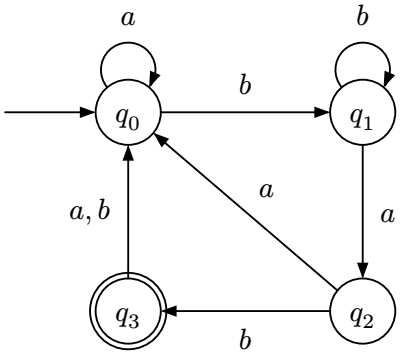


2. B = { w | w is a string over {a, b} and has both aa and ba as substrings }

Note: the string abbaa is an element of B.

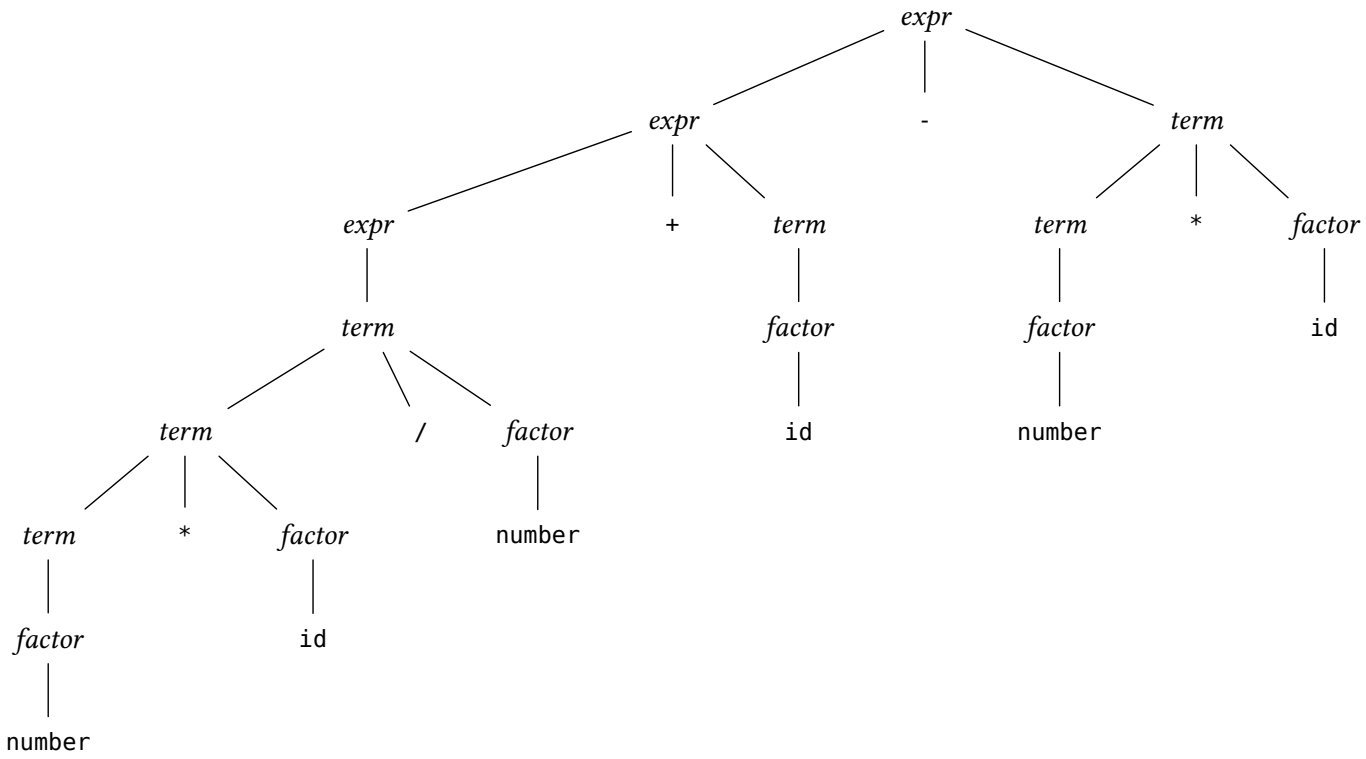


3. C = { w | w is a string over {a, b} and ends with bab }

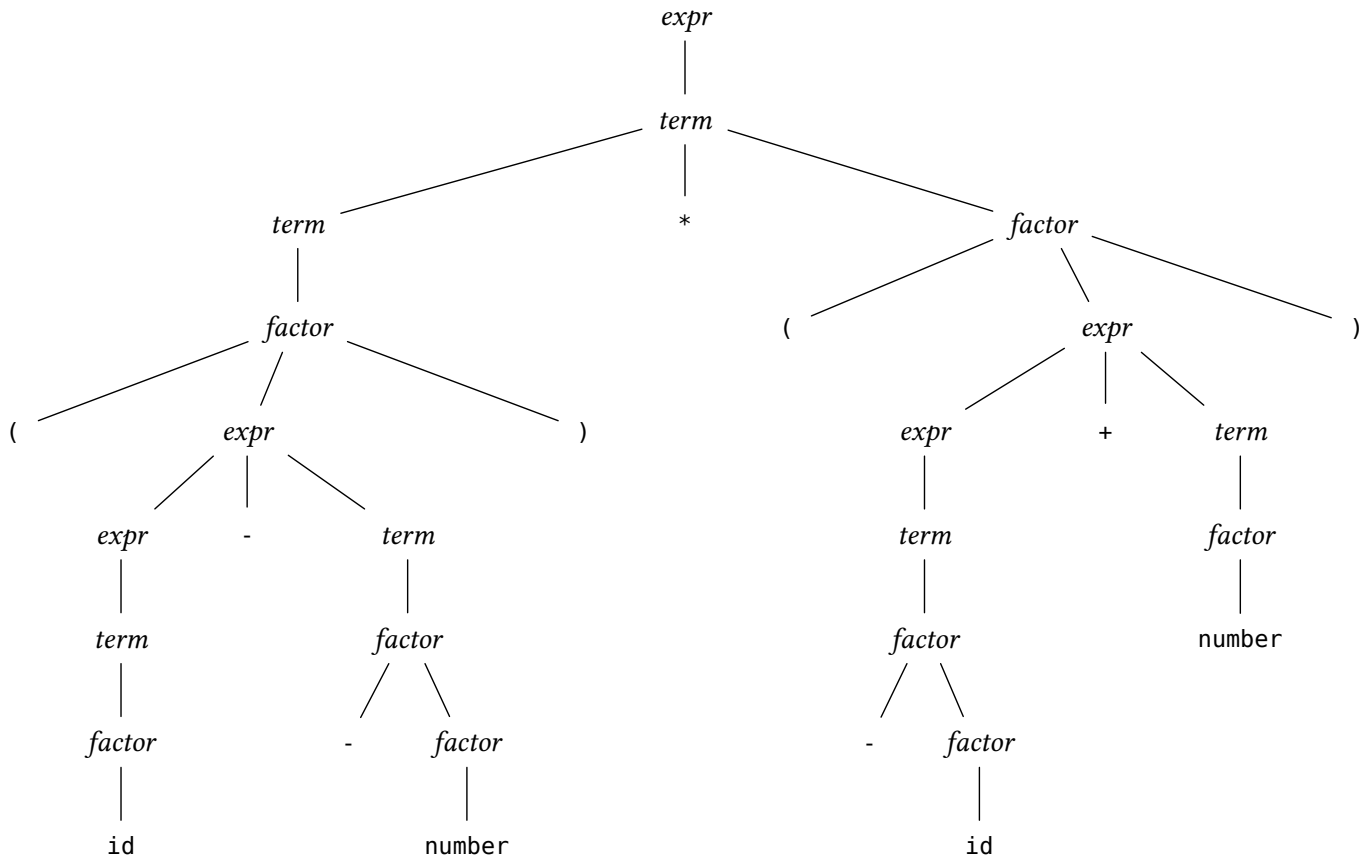


Problem 2 Context-Free Grammars: Derivations

1. number id / number + id - number id



2. (id - - number) ∗ (- id + number)



Problem 3: Context-Free Grammars: Create

1

S = aSb | A | B
A = aA | a
B = bB | b

2

S = aSb| aSa | bSa | bSb | a